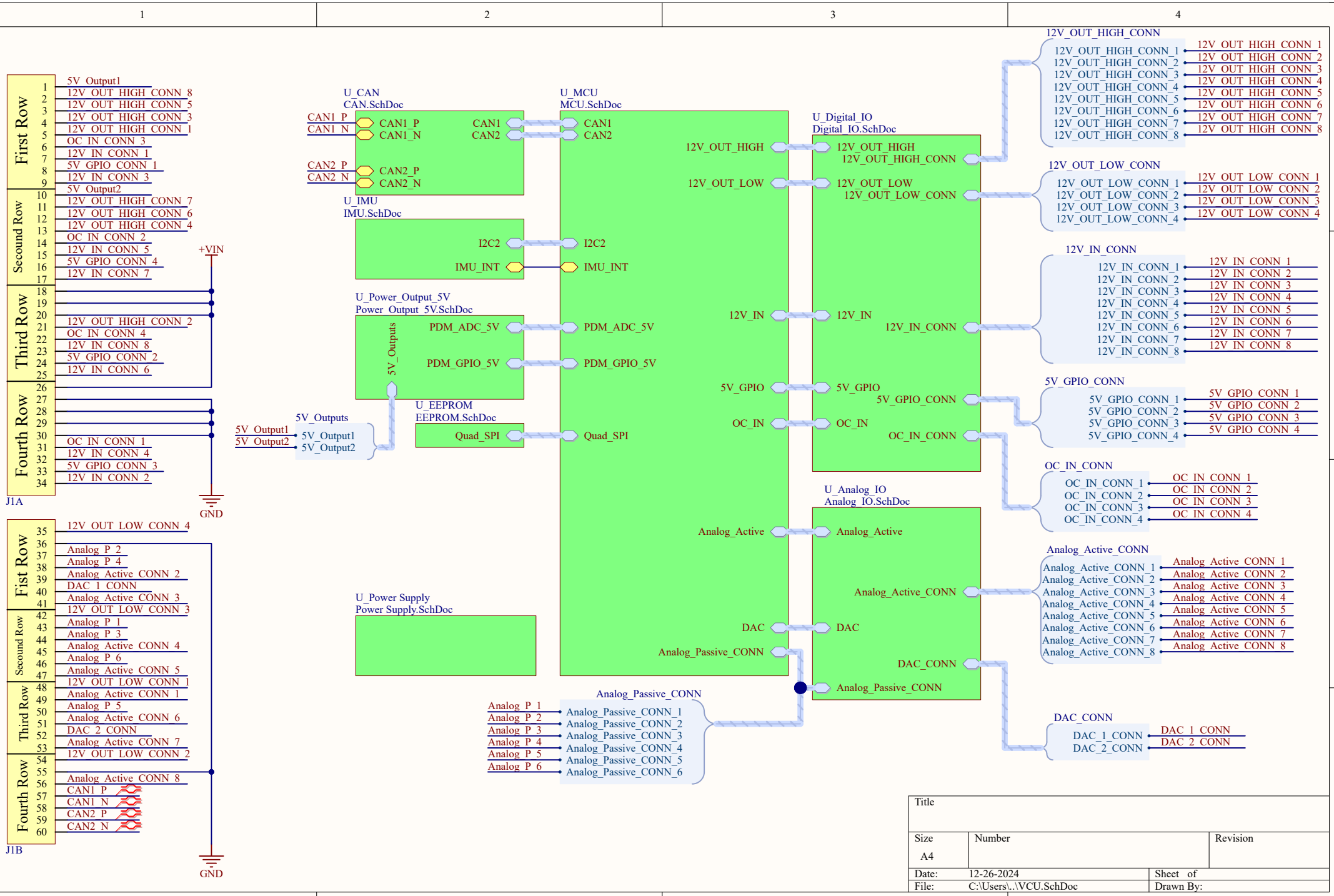
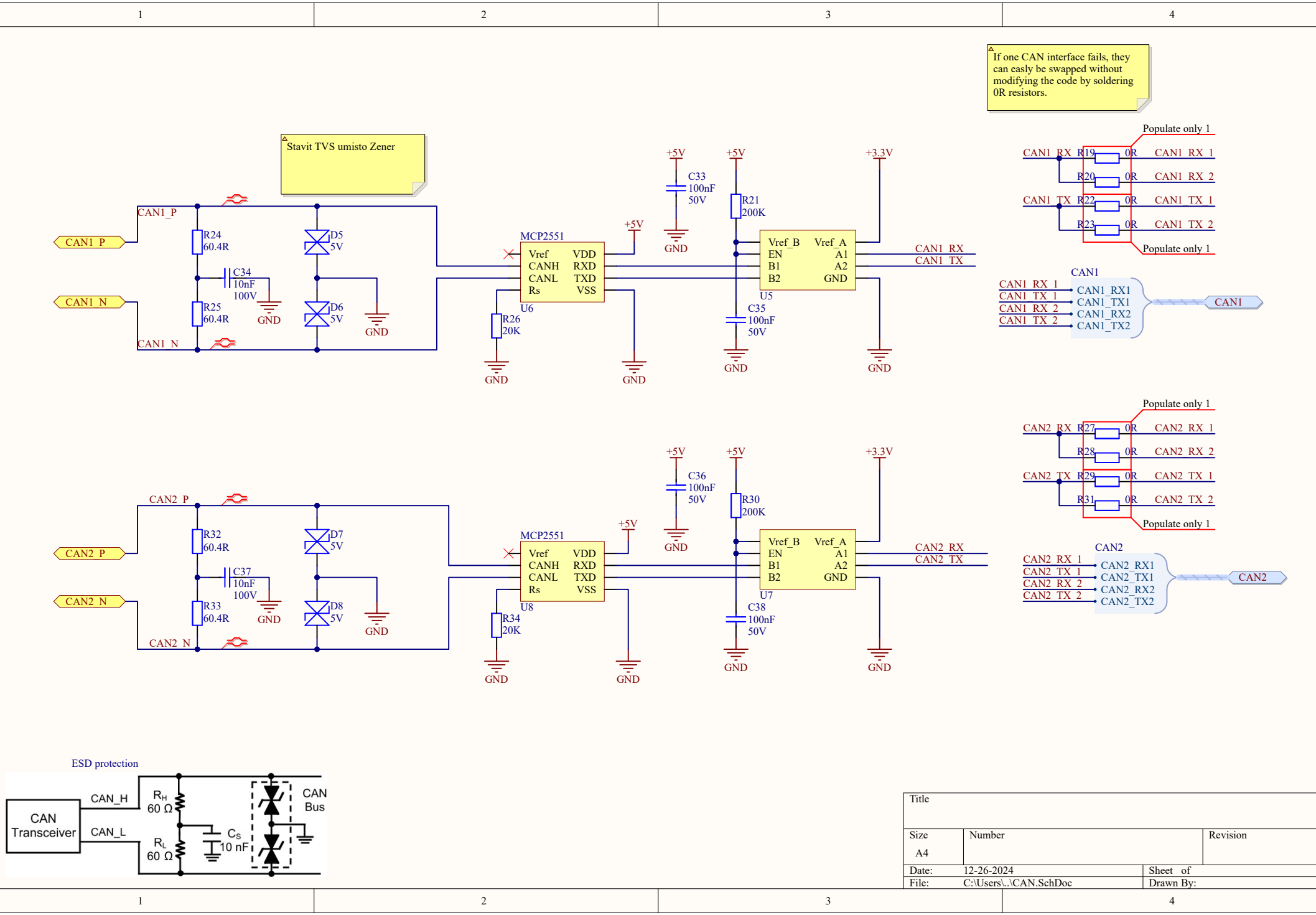
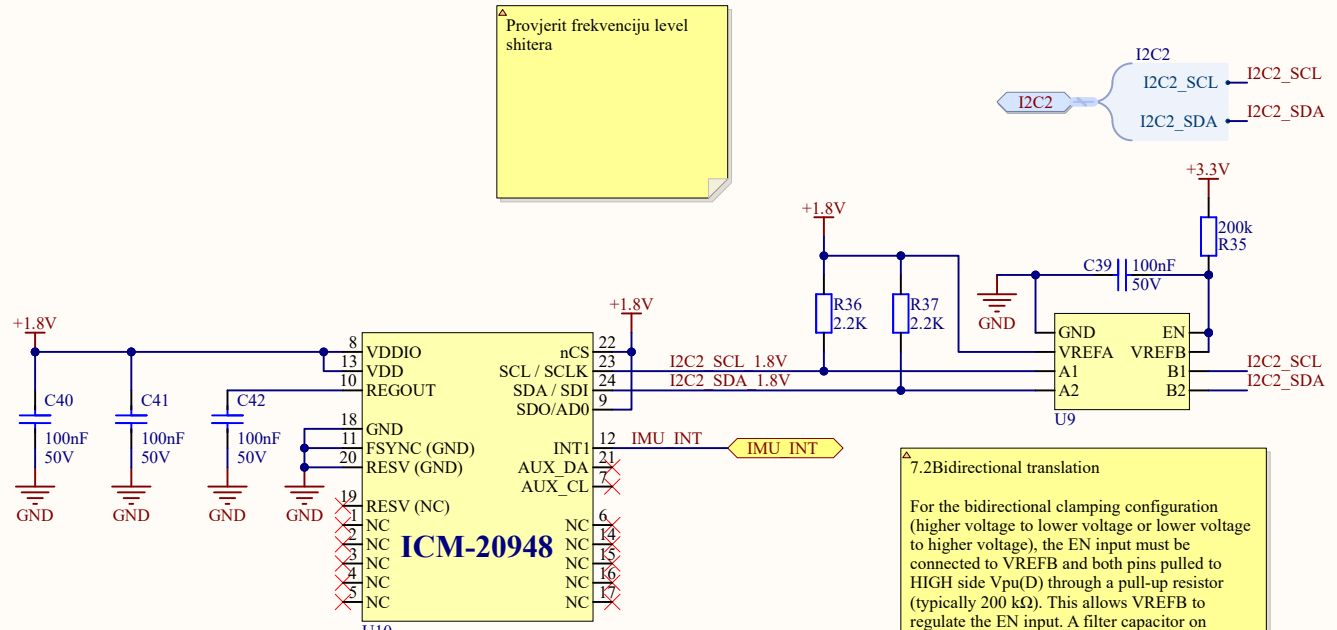

First Row
Secound Row
Third Row
Fourth Row
J1A
5V_Output1
12V_OUT_HIGH_CONN_8
12V_OUT_HIGH_CONN_5
12V_OUT_HIGH_CONN_3
12V_OUT_HIGH_CONN_1
OC_IN_CONN_3
12V_IN_CONN_1
5V_GPIO_CONN_1
12V_IN_CONN_3
5V_Output2
12V_OUT_HIGH_CONN_7
12V_OUT_HIGH_CONN_6
12V_OUT_HIGH_CONN_4
OC_IN_CONN_2
12V_IN_CONN_5
5V_GPIO_CONN_4
12V_IN_CONN_7
12V_OUT_HIGH_CONN_2
OC_IN_CONN_4
12V_IN_CONN_8
5V_GPIO_CONN_2
12V_IN_CONN_6
OC_IN_CONN_1
12V_IN_CONN_4
5V_GPIO_CONN_3
12V_IN_CONN_2
+VIN
GND
Fist Row
Secound Row
Third Row
Fourth Row
J1B
12V_OUT_LOW_CONN_4
Analog_P_2
Analog_P_4
Analog_Active_CONN_2
DAC_1_CONN
Analog_Active_CONN_3
12V_OUT_LOW_CONN_3
Analog_P_1
Analog_P_3
Analog_Active_CONN_4
Analog_P_6
Analog_Active_CONN_5
12V_OUT_LOW_CONN_1
Analog_Active_CONN_1
Analog_P_5
Analog_Active_CONN_6
DAC_2_CONN
Analog_Active_CONN_7
12V_OUT_LOW_CONN_2
Analog_Active_CONN_8
CAN1_P
CAN1_N
CAN2_P
CAN2_N
GND
U_CAN
CAN.SchDoc
CAN1_P
CAN1_N
CAN2_P
CAN2_N
CAN1
CAN2
U_IMU
IMU.SchDoc
I2C2
IMU_INT
U_Power_Output_5V
Power_Output_5V.SchDoc
5V_Outputs
PDM_ADC_5V
PDM_GPIO_5V
5V_Outputs
5V_Output1
5V_Output2
U_EEPROM
EEPROM.SchDoc
Quad_SPI
U_Power Supply
Power Supply.SchDoc
U_MCU
MCU.SchDoc
CAN1
CAN2
I2C2
IMU_INT
PDM_ADC_5V
PDM_GPIO_5V
Quad_SPI
12V_OUT_HIGH
12V_OUT_LOW
12V_IN
5V_GPIO
OC_IN
Analog_Active
DAC
Analog_Passive_CONN
U_Digital_IO
Digital_IO.SchDoc
12V_OUT_HIGH
12V_OUT_HIGH_CONN
12V_OUT_LOW
12V_OUT_LOW_CONN
12V_IN
12V_IN_CONN
5V_GPIO
5V_GPIO_CONN
OC_IN
OC_IN_CONN
U_Analog_IO
Analog_IO.SchDoc
Analog_Active
Analog_Active_CONN
DAC
DAC_CONN
Analog_Passive_CONN
Analog_Passive_CONN
Analog_P_1
Analog_P_2
Analog_P_3
Analog_P_4
Analog_P_5
Analog_P_6
Analog_Passive_CONN_1
Analog_Passive_CONN_2
Analog_Passive_CONN_3
Analog_Passive_CONN_4
Analog_Passive_CONN_5
Analog_Passive_CONN_6
12V_OUT_HIGH_CONN
12V_OUT_HIGH_CONN_1
12V_OUT_HIGH_CONN_2
12V_OUT_HIGH_CONN_3
12V_OUT_HIGH_CONN_4
12V_OUT_HIGH_CONN_5
12V_OUT_HIGH_CONN_6
12V_OUT_HIGH_CONN_7
12V_OUT_HIGH_CONN_8
12V_OUT_LOW_CONN
12V_OUT_LOW_CONN_1
12V_OUT_LOW_CONN_2
12V_OUT_LOW_CONN_3
12V_OUT_LOW_CONN_4
12V_IN_CONN
12V_IN_CONN_1
12V_IN_CONN_2
12V_IN_CONN_3
12V_IN_CONN_4
12V_IN_CONN_5
12V_IN_CONN_6
12V_IN_CONN_7
12V_IN_CONN_8
5V_GPIO_CONN
5V_GPIO_CONN_1
5V_GPIO_CONN_2
5V_GPIO_CONN_3
5V_GPIO_CONN_4
OC_IN_CONN
OC_IN_CONN_1
OC_IN_CONN_2
OC_IN_CONN_3
OC_IN_CONN_4
Analog_Active_CONN
Analog_Active_CONN_1
Analog_Active_CONN_2
Analog_Active_CONN_3
Analog_Active_CONN_4
Analog_Active_CONN_5
Analog_Active_CONN_6
Analog_Active_CONN_7
Analog_Active_CONN_8
DAC_CONN
DAC_1_CONN
DAC_2_CONN
Title
Size
A4
Number
Revision
Date:
12-26-2024
Sheet of
File:
C:\Users\..\VCU.SchDoc
Drawn By:


If one CAN interface fails, they can easly be swapped without modifying the code by soldering 0R resistors.
Stavit TVS umisto Zener
CAN1_P
CAN1_N
R24 60.4R
R25 60.4R
C34 10nF 100V
D5 5V
D6 5V
GND
MCP2551
Vref VDD
CANH RXD
CANL TXD
Rs VSS
U6
R26 20K
C33 100nF 50V
C35 100nF 50V
R21 200K
U5
Vref_B Vref_A
EN A1
B1 A2
B2 GND
+5V
+3.3V
CAN1_RX
CAN1_TX
Populate only 1
CAN1_RX_R19 0R CAN1_RX_1
R20 0R CAN1_RX_2
CAN1_TX_R22 0R CAN1_TX_1
R23 0R CAN1_TX_2
CAN1
CAN1_RX_1
CAN1_TX_1
CAN1_RX_2
CAN1_TX_2
CAN1_RX1
CAN1_TX1
CAN1_RX2
CAN1_TX2
CAN2_P
CAN2_N
R32 60.4R
R33 60.4R
C37 10nF 100V
D7 5V
D8 5V
U8
R34 20K
C36 100nF 50V
C38 100nF 50V
R30 200K
U7
CAN2_RX
CAN2_TX
CAN2_RX_R27 0R CAN2_RX_1
R28 0R CAN2_RX_2
CAN2_TX_R29 0R CAN2_TX_1
R31 0R CAN2_TX_2
CAN2
CAN2_RX_1
CAN2_TX_1
CAN2_RX_2
CAN2_TX_2
CAN2_RX1
CAN2_TX1
CAN2_RX2
CAN2_TX2
ESD protection
CAN Transceiver
CAN_H
CAN_L
R_H 60 Ω
R_L 60 Ω
C_S 10 nF
CAN Bus
Title
Size A4
Number
Revision
Date: 12-26-2024
Sheet of
File: C:\Users\..\CAN.SchDoc
Drawn By:



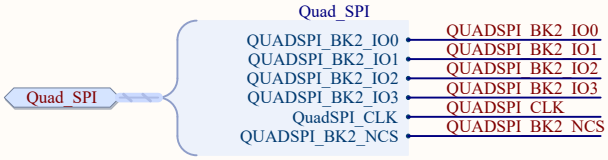
7.2Bidirectional translation

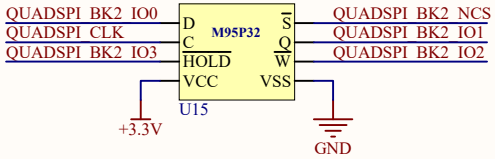
For the bidirectional clamping configuration (higher voltage to lower voltage or lower voltage to higher voltage), the EN input must be connected to VREFB and both pins pulled to HIGH side Vpu(D) through a pull-up resistor (typically 200 kΩ). This allows VREFB to regulate the EN input. A filter capacitor on VREFB is recommended.

4.4 EXPOSED DIE PAD PRECAUTIONS

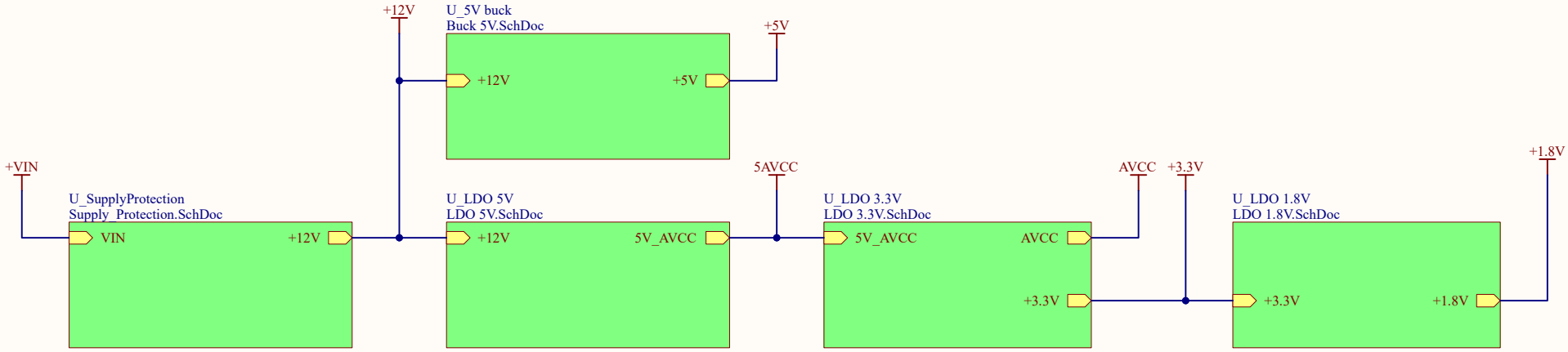
InvenSense products have very low active and standby current consumption. The exposed die pad is not required for heat sinking, and should not be soldered to the PCB. Failure to adhere to this rule can induce performance changes due to package thermo-mechanical stress. There is no electrical connection between the pad and the CMOS.

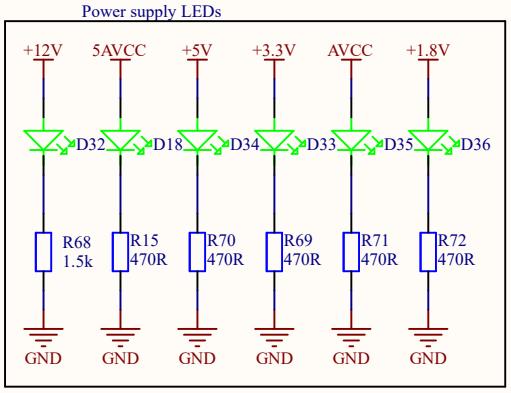
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Size: A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\IMU.SchDoc		Drawn By:





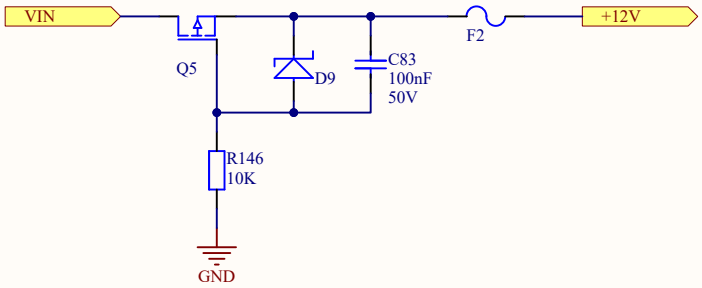
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Date: 12-26-2024		Sheet of
File: C:\Users\..\EEPROM.SchDoc		Drawn By:



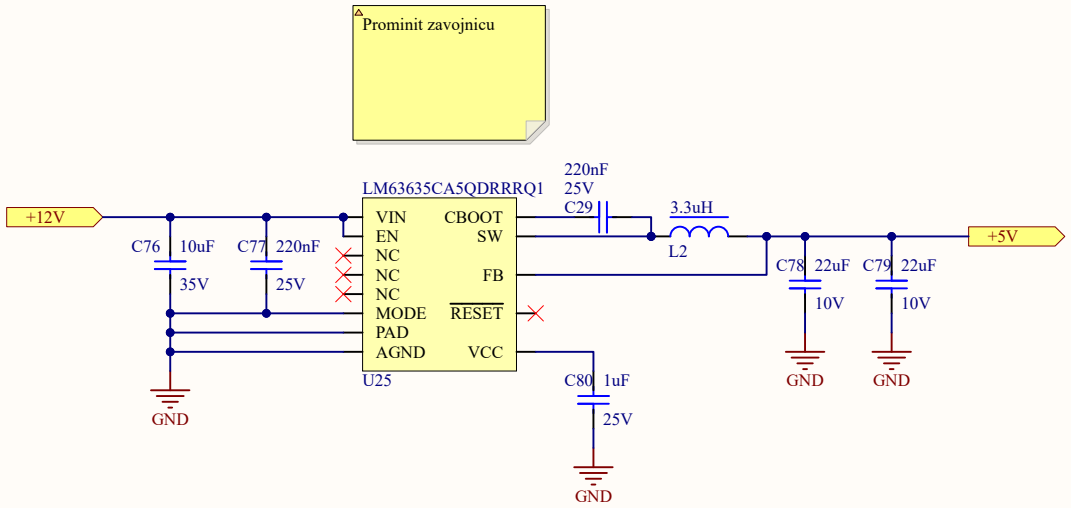


Treba li otpornik na 1.8V

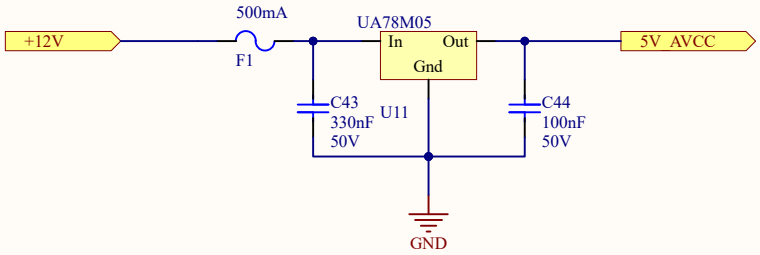
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Date: 12-26-2024		Sheet of
File: C:\Users\..\Power Supply.SchDoc		Drawn By:



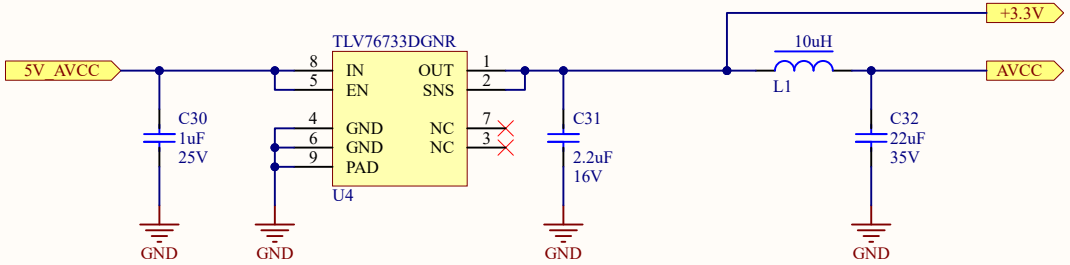
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Size A	Number		Revision
Date:	12-26-2024	Sheet of	
File:	C:\Users\..\Supply_Protection.SchDoc	Drawn By:	



Title		
Size: A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\Buck 5V.SchDoc		Drawn By:

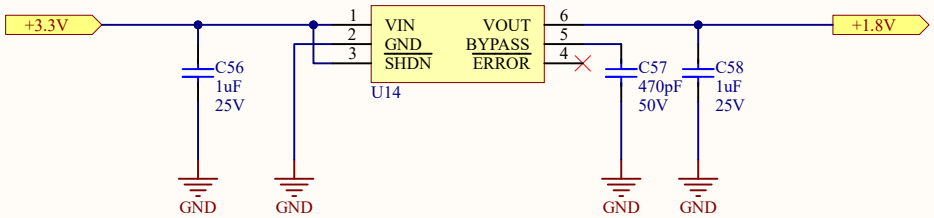


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Size: A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\LDO 5V.SchDoc		Drawn By:

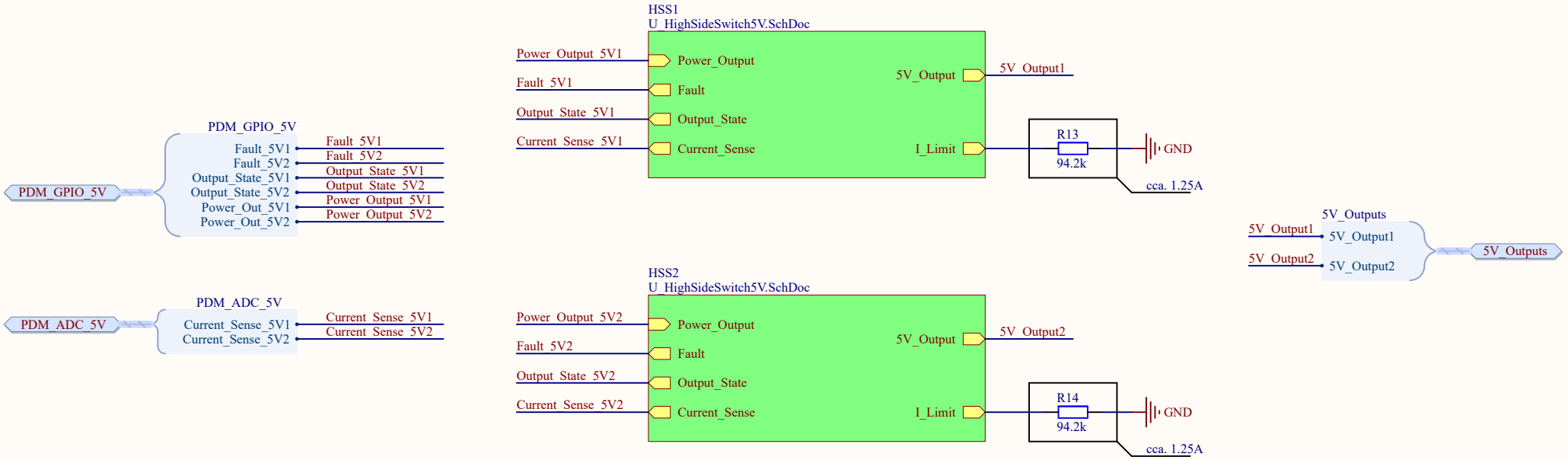


Title			
Size A4	Number		Revision
Date:	12-26-2024	Sheet of	
File:	C:\Users\..\LDO 3.3V.SchDoc	Drawn By:	

footprint za regulator - mech 7 i provjerit raspored pinova



Title		
Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\LDO 1.8V.SchDoc		Drawn By:



Title		
Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\Power_Output_5V.SchDoc		Drawn By:

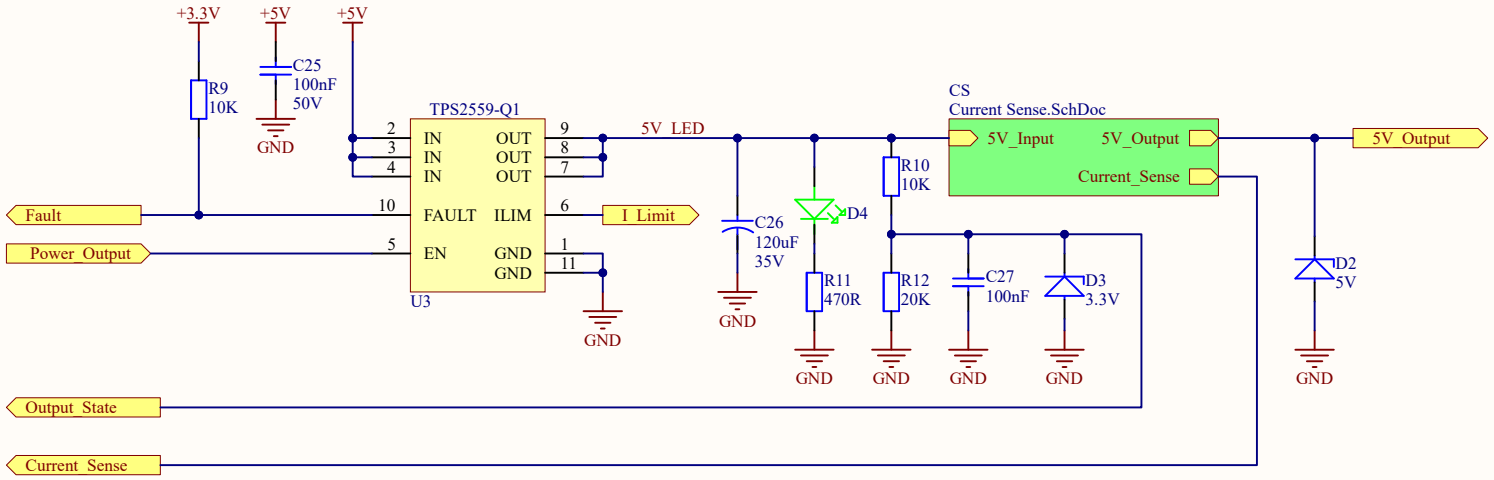
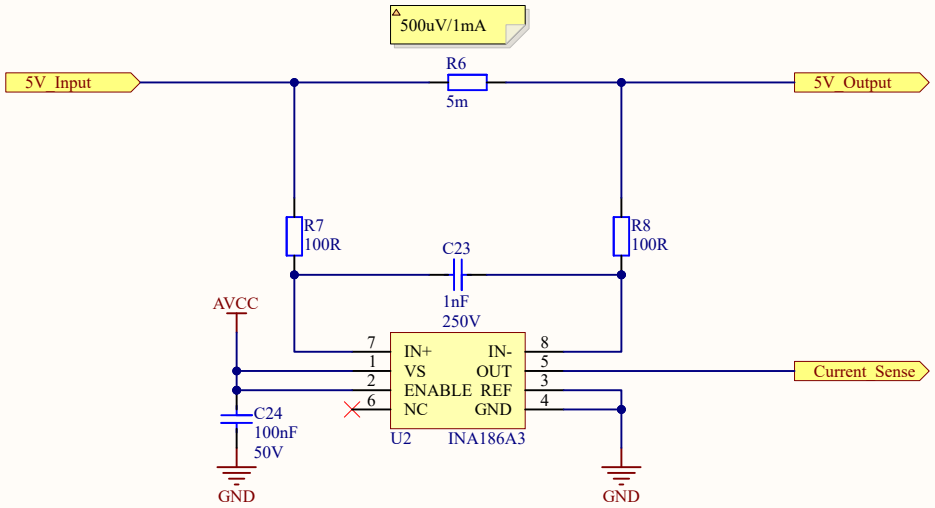


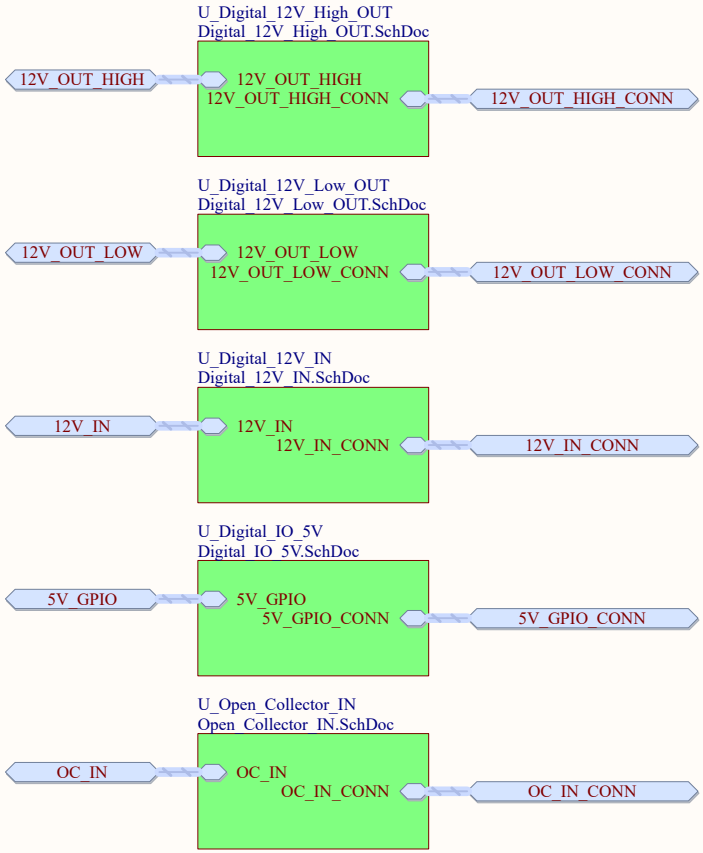
Table 1. Common $R_{(ILIM)}$ Resistor Selections

DESIRED NOMINAL CURRENT LIMIT (mA)	IDEAL RESISTOR (kΩ)	CLOSEST 1% RESISTOR (kΩ)	RESISTOR TOLERANCE		ACTUAL LIMITS		
			1% LOW (kΩ)	1% HIGH (kΩ)	I_{OS} MIN (mA)	I_{OS} NOM (mA)	I_{OS} MAX (mA)
1250	94.1	93.1	92.2	94	1152.7	1263.7	1368.2
1500	78.4	78.7	77.9	79.5	1372.5	1495.1	1610.9
1750	67.2	66.5	65.8	67.2	1633.2	1769.7	1893.3
2000	58.8	59	58.4	59.6	1847	1994.8	2133.7
2250	52.3	52.3	51.8	52.8	2089.9	2550.6	2400.9
2500	47.1	47.5	47	48	2306	2478.2	2638.4
2750	42.8	43.2	42.8	43.6	2540.5	2725.1	2895.8
3000	39.2	39.2	38.8	39.6	2804.8	3003.4	3185.7
3250	36.2	36.5	36.1	36.9	3016	3225.7	3417.2
3500	33.6	34	33.7	34.3	3241.4	3463.1	3664.1
3750	31.4	31.6	31.3	31.9	3491.5	3726.4	3937.8
4000	29.4	29.4	29.1	29.7	3756.5	4005.4	4227.7
4250	27.7	28	27.7	28.3	3946.9	4205.9	4435.8
4500	26.2	26.1	25.8	26.4	4237.9	4512.3	4753.9
4750	24.8	24.9	24.7	25.1	4444.6	4729.9	4979.6

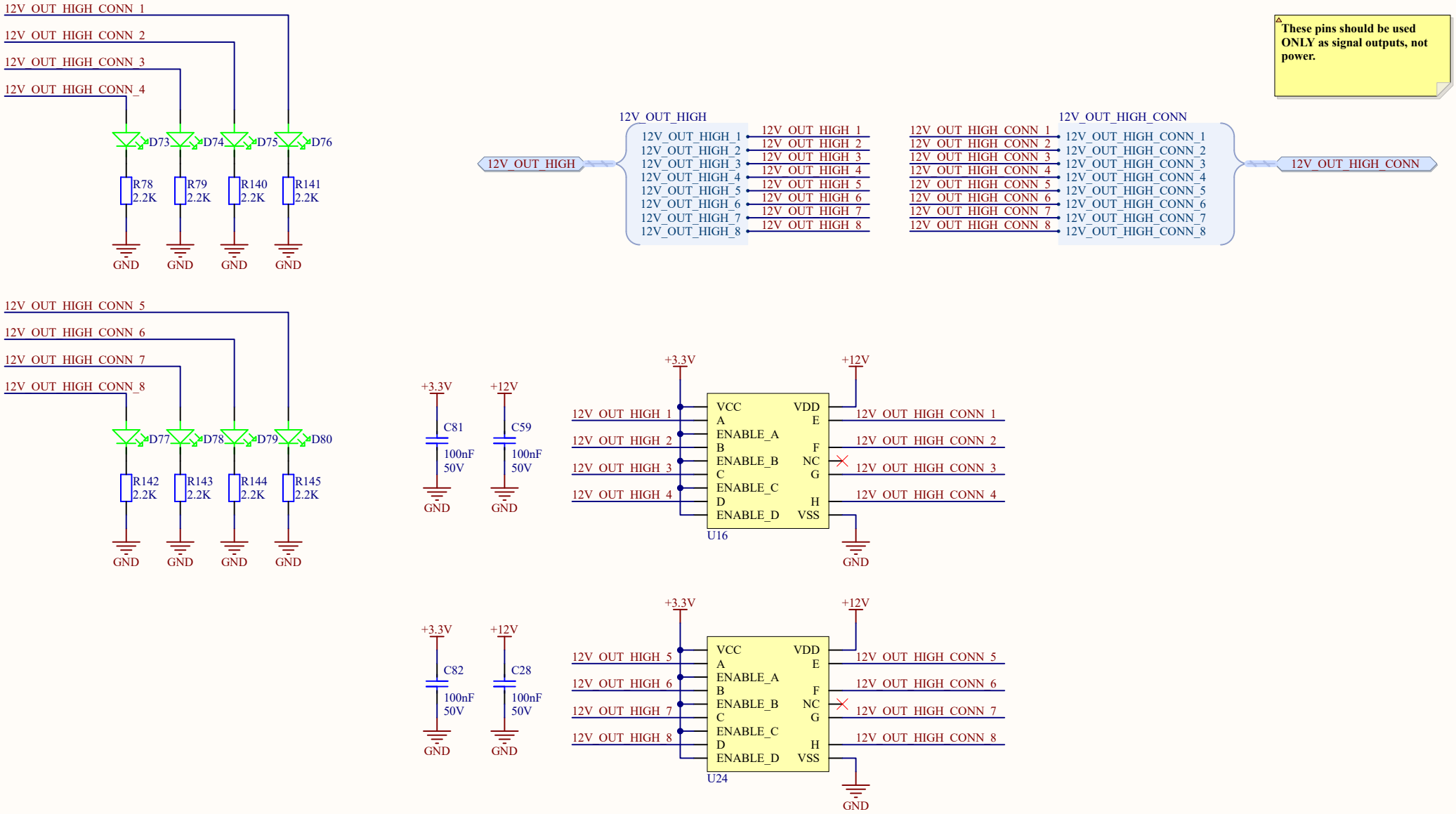
Title			
Size A4	Number		Revision
Date:	12-26-2024	Sheet of	
File:	C:\Users\..\U_HighSideSwitch5V.SchDoc	Drawn By:	



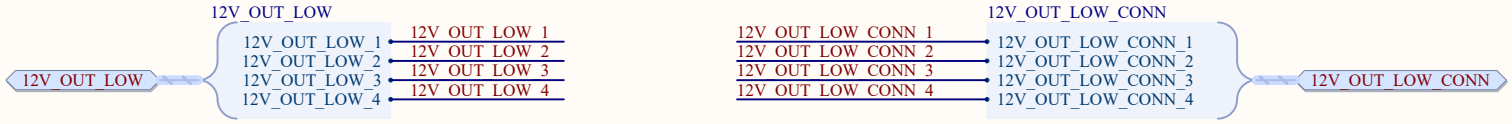
Title			
Size A4	Number		Revision
Date:	12-26-2024	Sheet of	
File:	C:\Users\..\Current Sense.SchDoc	Drawn By:	



Title			
Size A4	Number		Revision
Date:	12-26-2024	Sheet of	
File:	C:\Users\..\Digital_IO.SchDoc	Drawn By:	



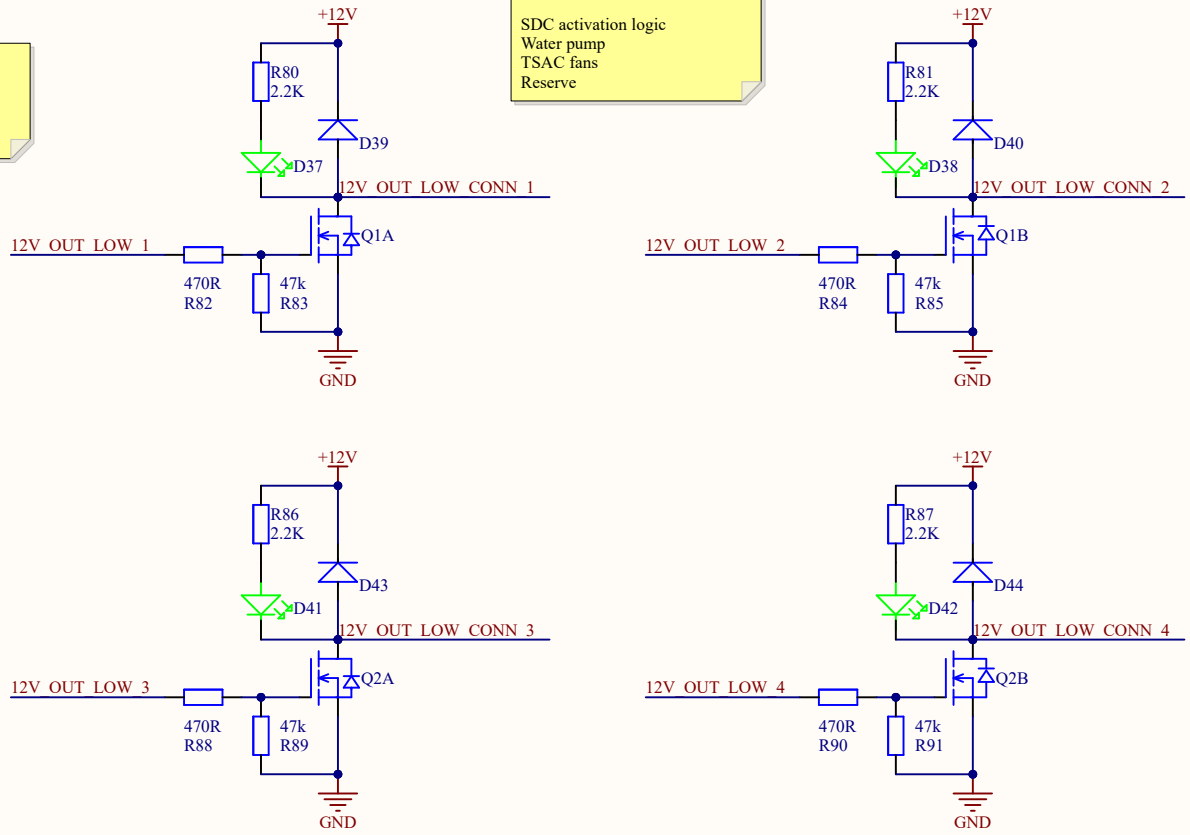
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Date: 12-26-2024	Sheet of	
File: C:\Users\..\Digital_12V_High_OUT.SchDoc	Drawn By:	



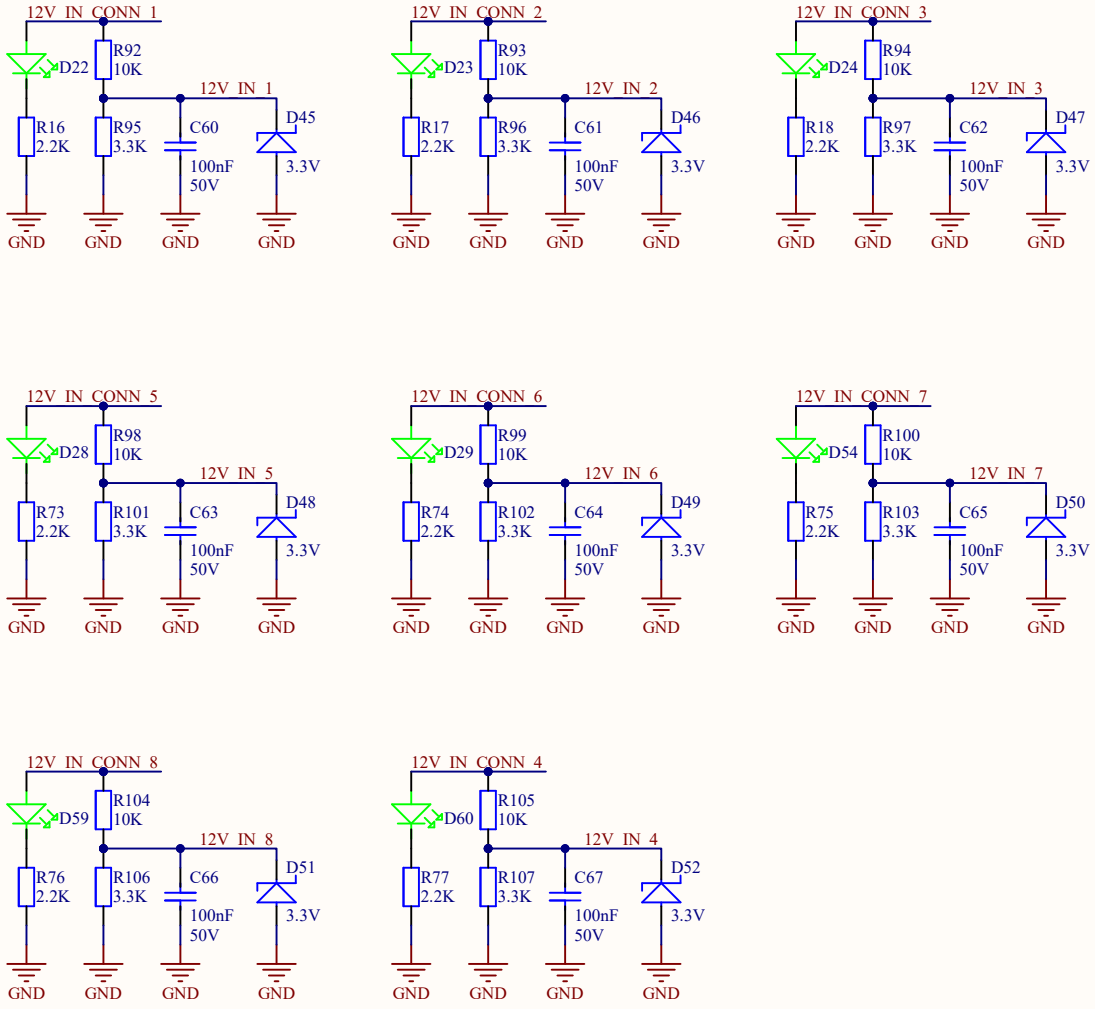
Dodat mosfet na git

Primary use of these outputs is to drive relays.

SDC activation logic
Water pump
TSAC fans
Reserve

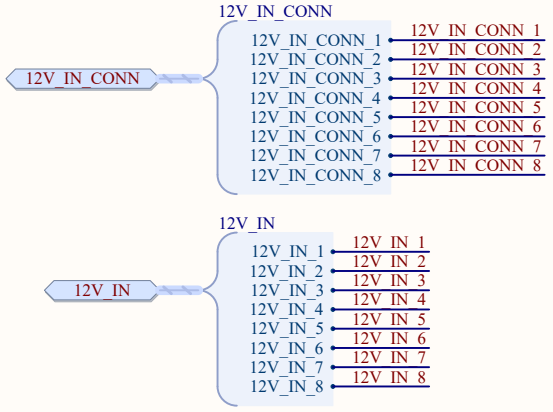


Title		
Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\Digital_12V_Low_OUT.SchDoc		Drawn By:

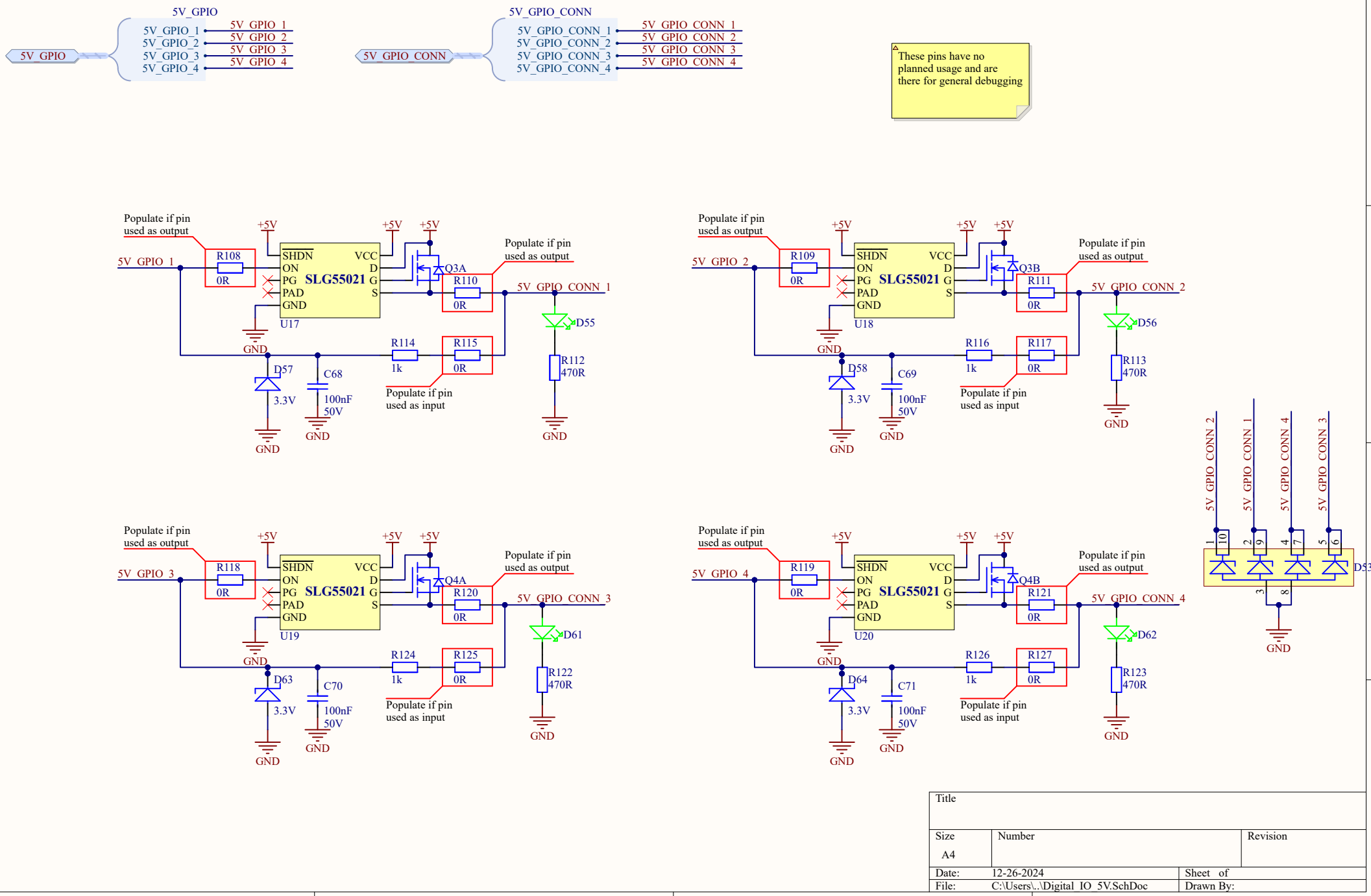


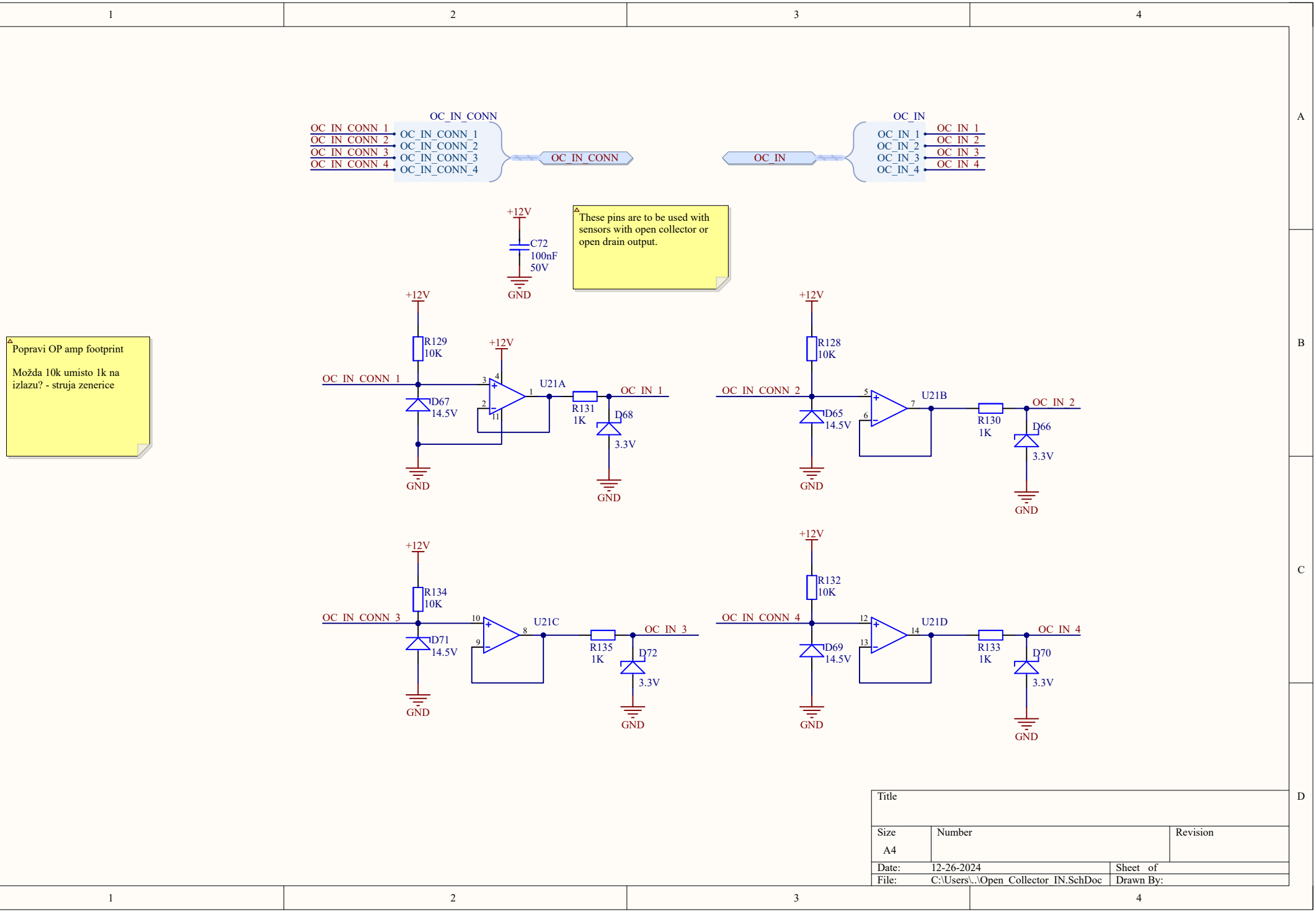
Primary use of these pins is to track the state of each SDC element. Those are:

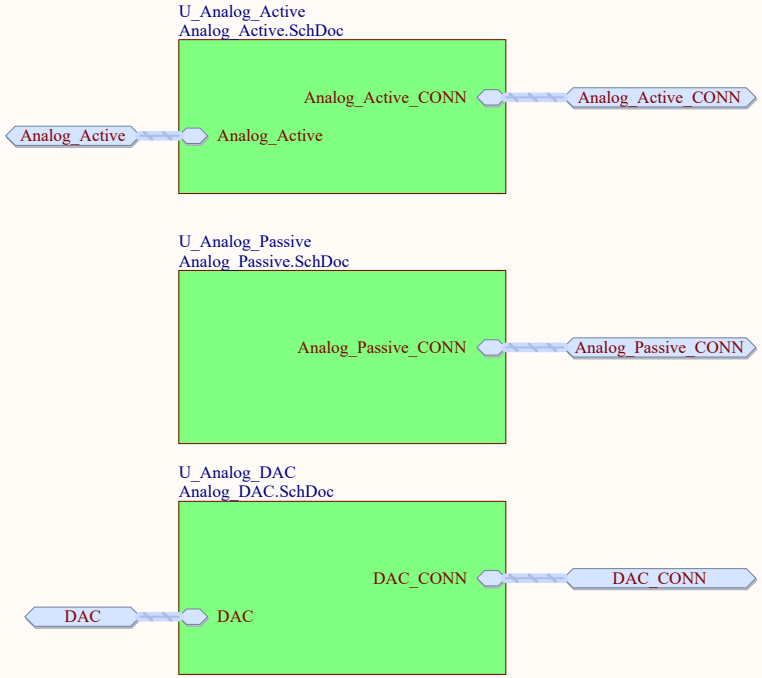
BSPD relay
Shutdown button - L
Shutdown button - R
Shutdown button - Cockpit
BOTS
TSMS
Reserve 1
Reserve 2



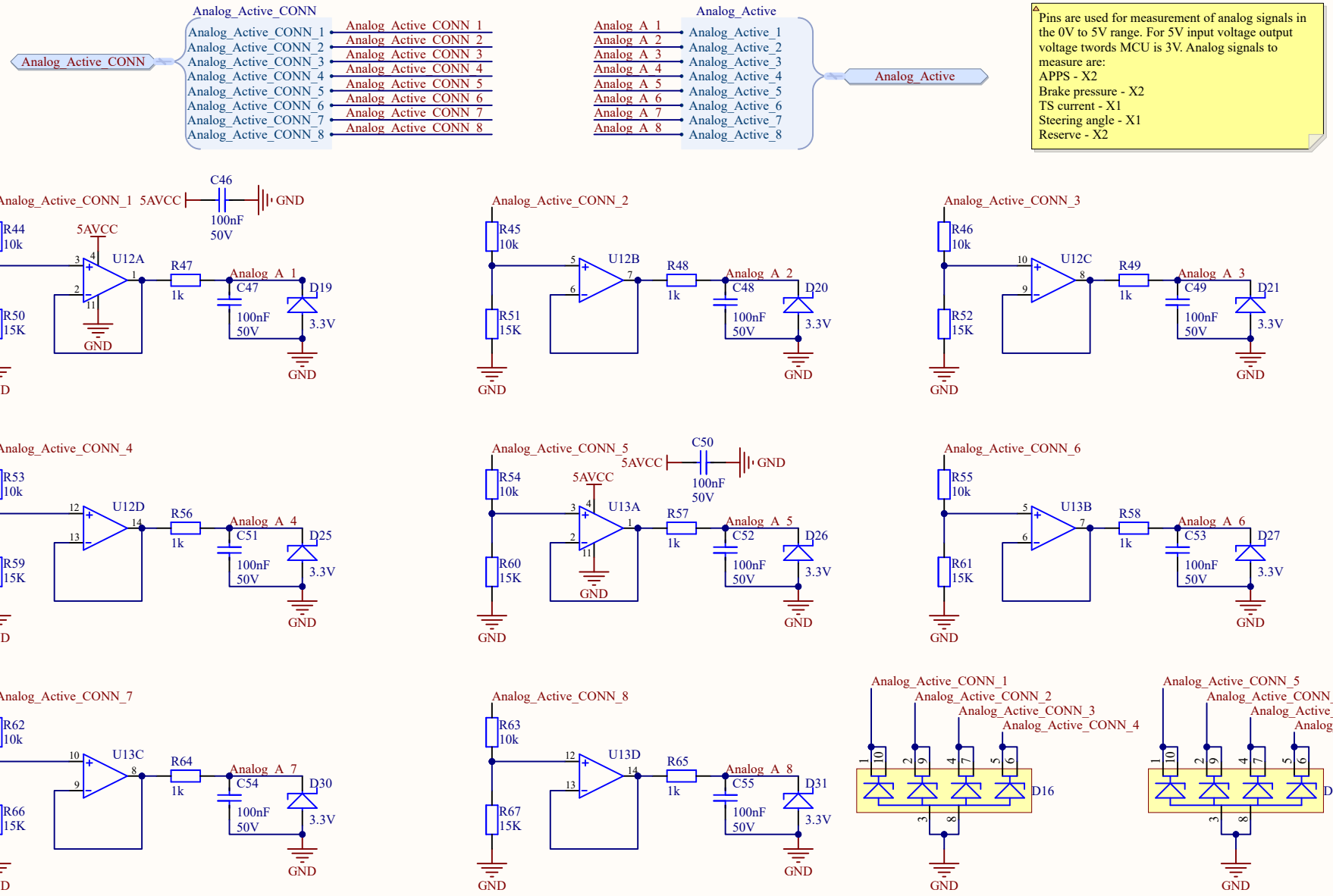
Title		
Size: A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\Digital_12V_IN.SchDoc		Drawn By:

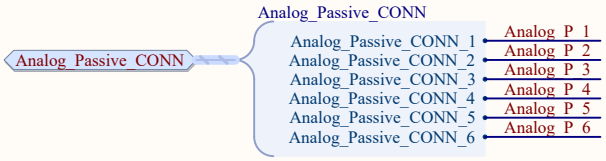

5V_GPIO
5V_GPIO_1
5V_GPIO_2
5V_GPIO_3
5V_GPIO_4
5V_GPIO_1
5V_GPIO_2
5V_GPIO_3
5V_GPIO_4
5V_GPIO_CONN
5V_GPIO_CONN_1
5V_GPIO_CONN_2
5V_GPIO_CONN_3
5V_GPIO_CONN_4
5V_GPIO_CONN_1
5V_GPIO_CONN_2
5V_GPIO_CONN_3
5V_GPIO_CONN_4
These pins have no planned usage and are there for general debugging
Populate if pin used as output
Populate if pin used as input
5V_GPIO_1
5V_GPIO_2
5V_GPIO_3
5V_GPIO_4
5V_GPIO_CONN_1
5V_GPIO_CONN_2
5V_GPIO_CONN_3
5V_GPIO_CONN_4
R108
R109
R118
R119
R110
R111
R120
R121
R114
R115
R116
R117
R124
R125
R126
R127
R112
R113
R122
R123
0R
1k
470R
SLG55021
SHDN
ON
PG
PAD
GND
VCC
D
G
S
U17
U18
U19
U20
Q3A
Q3B
Q4A
Q4B
D55
D56
D61
D62
D57
D58
D63
D64
3.3V
C68
C69
C70
C71
100nF
50V
+5V
GND
D53
5V_GPIO_CONN_2
5V_GPIO_CONN_1
5V_GPIO_CONN_4
5V_GPIO_CONN_3
1
10
2
9
4
7
5
6
3
8
Title
Size
A4
Number
Revision
Date:
12-26-2024
Sheet of
File:
C:\Users\..\Digital_IO_5V.SchDoc
Drawn By:


OC_IN_CONN
OC_IN_CONN_1
OC_IN_CONN_2
OC_IN_CONN_3
OC_IN_CONN_4
OC_IN_CONN
OC_IN
OC_IN_1
OC_IN_2
OC_IN_3
OC_IN_4
+12V
C72
100nF
50V
GND
These pins are to be used with sensors with open collector or open drain output.
Popravi OP amp footprint
Možda 10k umisto 1k na izlazu? - struja zenerice
R129
10K
R131
1K
D67
14.5V
D68
3.3V
U21A
R128
10K
R130
1K
D65
D66
U21B
R134
10K
R135
1K
D71
D72
U21C
R132
10K
R133
1K
D69
D70
U21D
Title
Size
A4
Number
Revision
Date:
12-26-2024
Sheet of
File:
C:\Users\..\Open Collector IN.SchDoc
Drawn By:

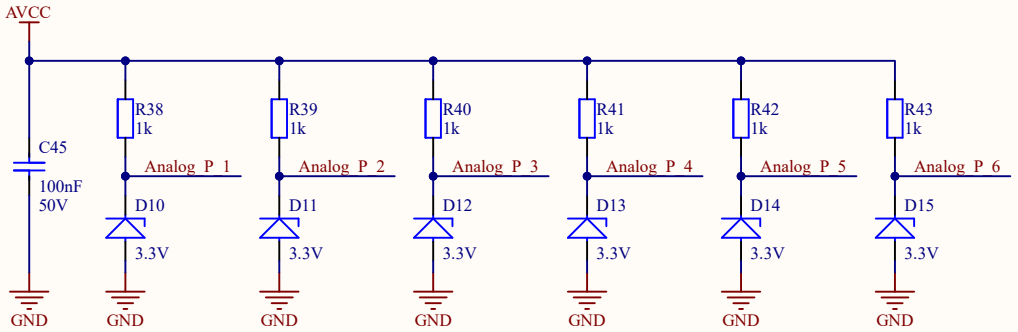


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Date: 12-26-2024		Sheet of
File: C:\Users\..\Analog_IO.SchDoc		Drawn By:

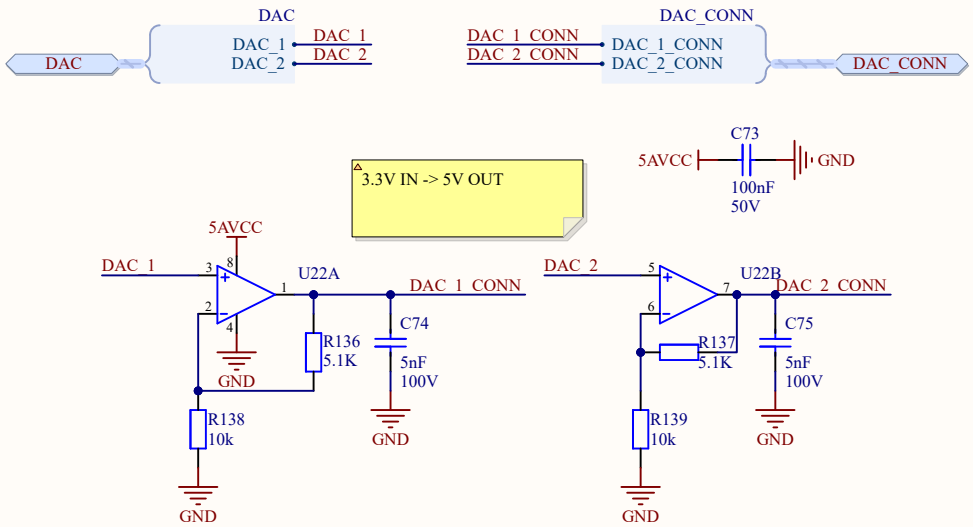

Analog_Active_CONN
Analog_Active_CONN
Analog_Active_CONN_1
Analog_Active_CONN_2
Analog_Active_CONN_3
Analog_Active_CONN_4
Analog_Active_CONN_5
Analog_Active_CONN_6
Analog_Active_CONN_7
Analog_Active_CONN_8
Analog_Active_CONN_1
Analog_Active_CONN_2
Analog_Active_CONN_3
Analog_Active_CONN_4
Analog_Active_CONN_5
Analog_Active_CONN_6
Analog_Active_CONN_7
Analog_Active_CONN_8
Analog_Active
Analog_A_1
Analog_A_2
Analog_A_3
Analog_A_4
Analog_A_5
Analog_A_6
Analog_A_7
Analog_A_8
Analog_Active_1
Analog_Active_2
Analog_Active_3
Analog_Active_4
Analog_Active_5
Analog_Active_6
Analog_Active_7
Analog_Active_8
Analog_Active
Pins are used for measurement of analog signals in the 0V to 5V range. For 5V input voltage output voltage twords MCU is 3V. Analog signals to measure are:
APPS - X2
Brake pressure - X2
TS current - X1
Steering angle - X1
Reserve - X2
Analog_Active_CONN_1
5AVCC
C46
100nF
50V
GND
R44
10k
R50
15K
U12A
R47
1k
Analog_A_1
C47
100nF
50V
D19
3.3V
Analog_Active_CONN_2
R45
10k
R51
15K
U12B
R48
1k
Analog_A_2
C48
100nF
50V
D20
3.3V
Analog_Active_CONN_3
R46
10k
R52
15K
U12C
R49
1k
Analog_A_3
C49
100nF
50V
D21
3.3V
Analog_Active_CONN_4
R53
10k
R59
15K
U12D
R56
1k
Analog_A_4
C51
100nF
50V
D25
3.3V
Analog_Active_CONN_5
C50
5AVCC
100nF
50V
R54
10k
R60
15K
U13A
R57
1k
Analog_A_5
C52
100nF
50V
D26
3.3V
Analog_Active_CONN_6
R55
10k
R61
15K
U13B
R58
1k
Analog_A_6
C53
100nF
50V
D27
3.3V
Analog_Active_CONN_7
R62
10k
R66
15K
U13C
R64
1k
Analog_A_7
C54
100nF
50V
D30
3.3V
Analog_Active_CONN_8
R63
10k
R67
15K
U13D
R65
1k
Analog_A_8
C55
100nF
50V
D31
3.3V
D16
D17
Title
Size
A4
Number
Revision
Date:
12-26-2024
Sheet of
File:
C:\Users\..\Analog_Active.SchDoc
Drawn By:



Pins are used with sensors that change resistance such as thermistors. Planned usage is:
Thermistor - X2
Suspension position sensor - X4



Title		
Size: A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\Analog_Passive.SchDoc		Drawn By:

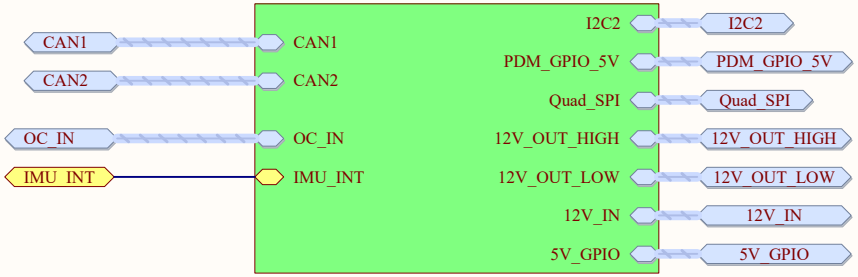


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Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\Analog_DAC.SchDoc		Drawn By:

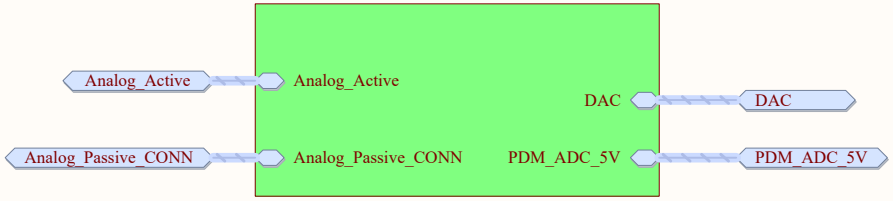
U_MCU System
MCU System.SchDoc

U_MCU Power
MCU Power.SchDoc

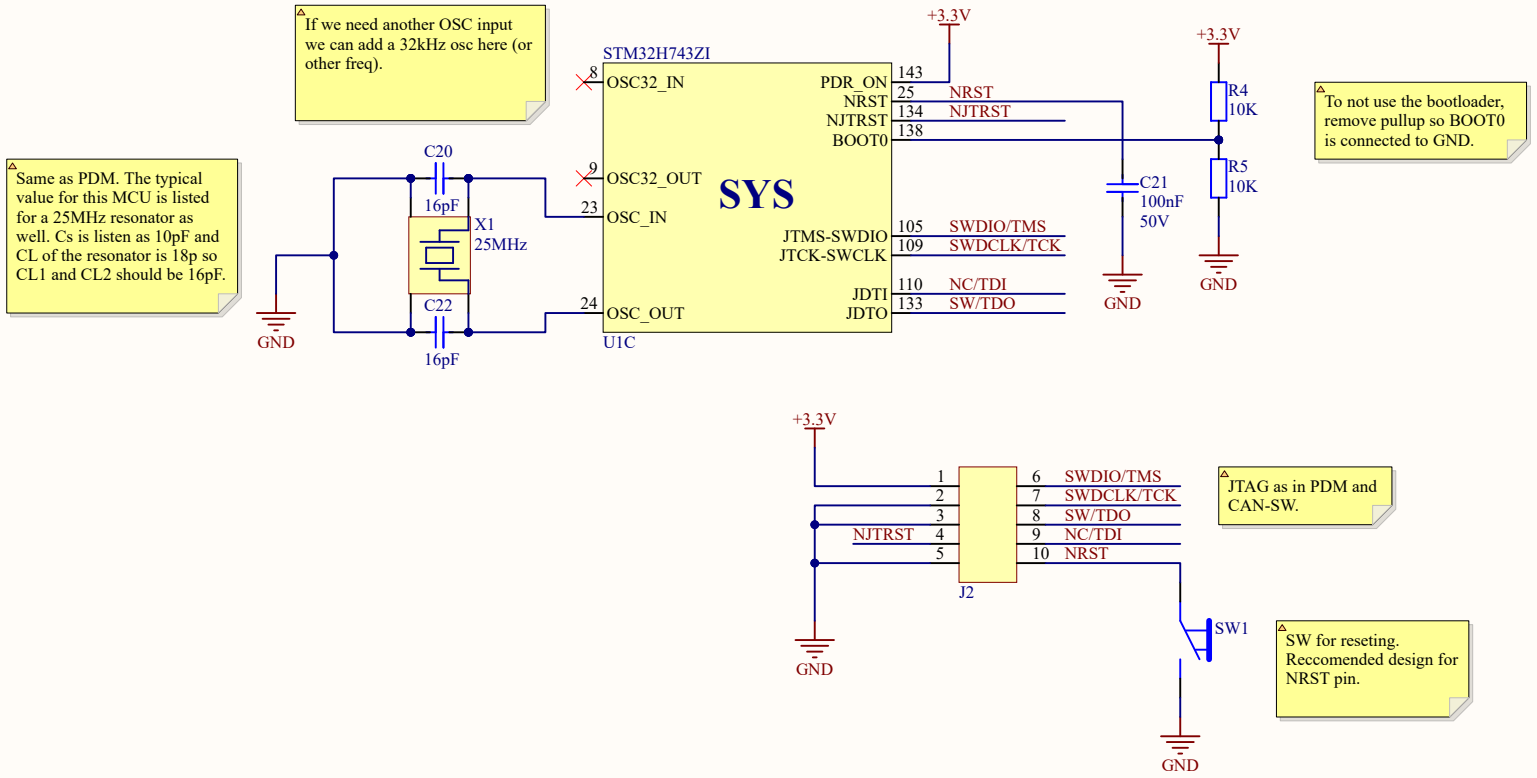
U_MCU GPIO
MCU GPIO.SchDoc



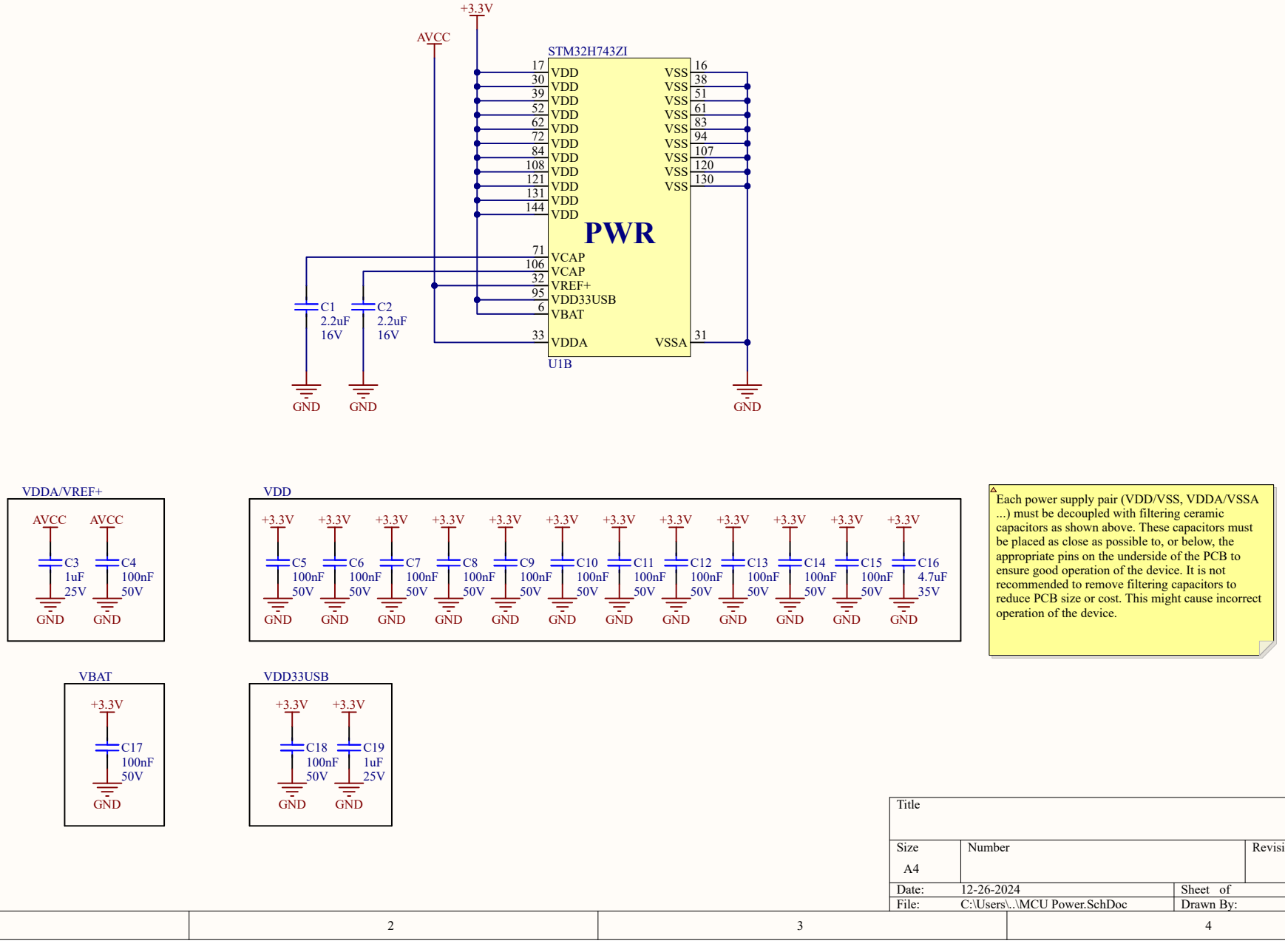
U_MCU ADC
MCU ADC.SchDoc

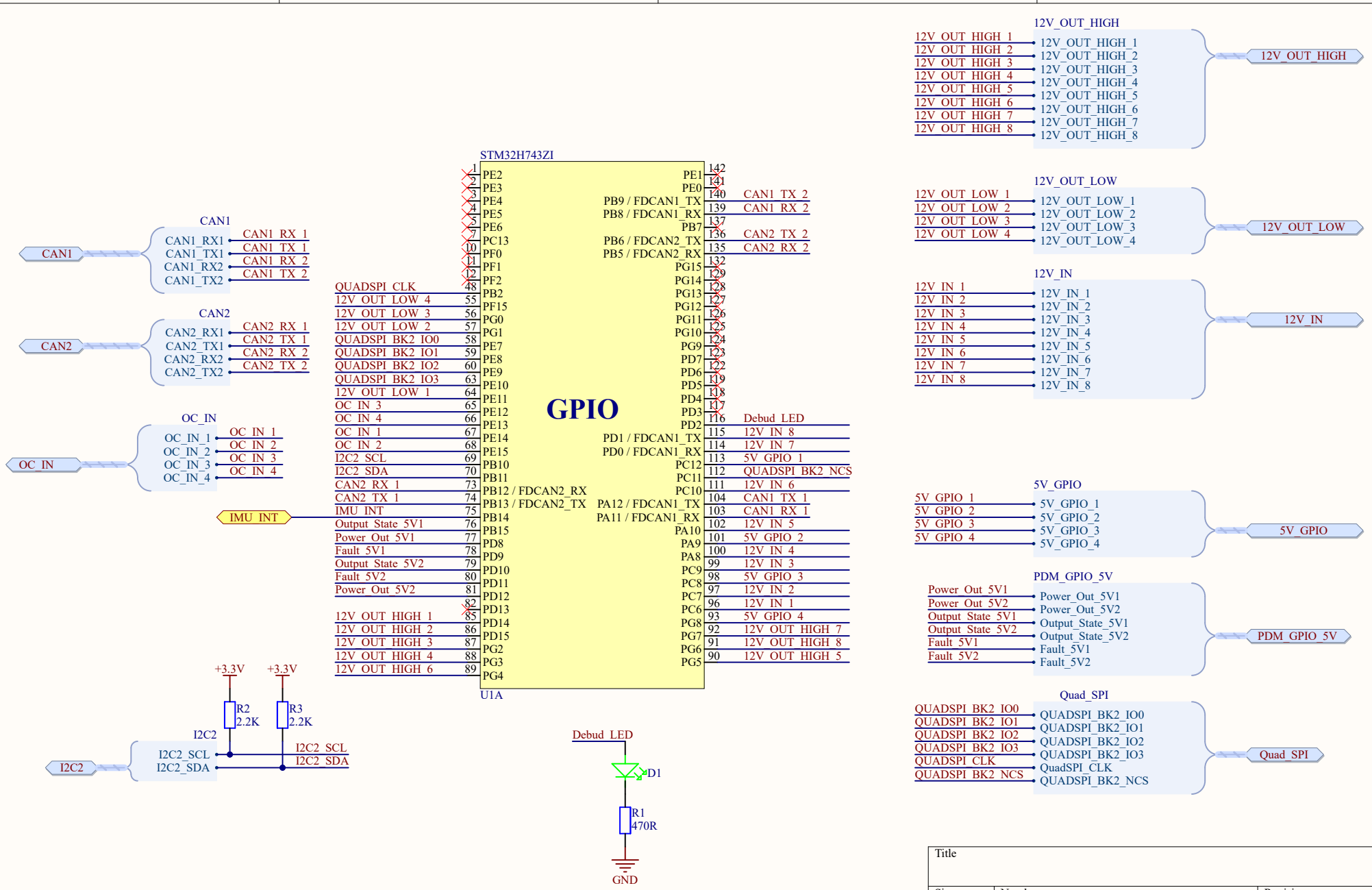


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Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\MCU.SchDoc		Drawn By:

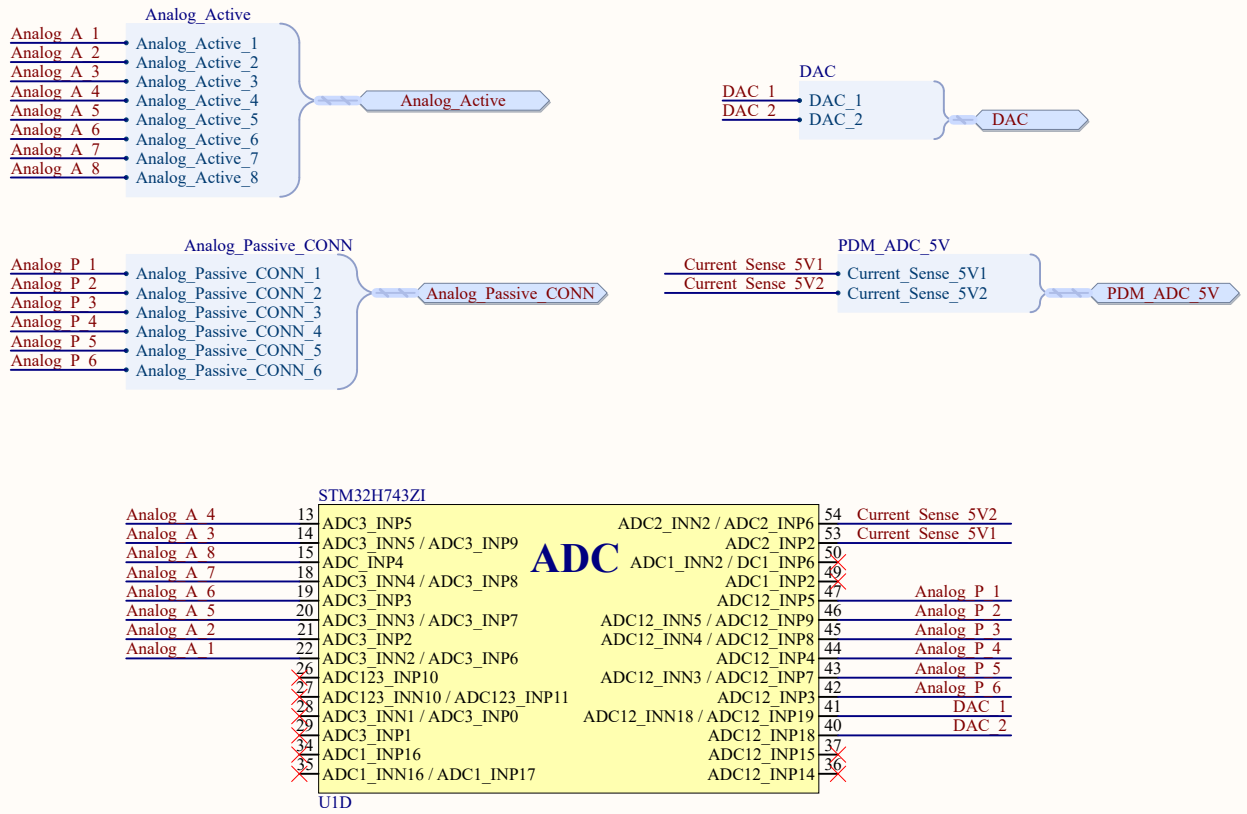


Title		
Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\MCU System.SchDoc		Drawn By:


+3.3V
AVCC
STM32H743ZI
17 VDD
30 VDD
39 VDD
52 VDD
62 VDD
72 VDD
84 VDD
108 VDD
121 VDD
131 VDD
144 VDD
PWR
71 VCAP
106 VCAP
32 VREF+
95 VDD33USB
6 VBAT
33 VDDA
VSS 16
VSS 38
VSS 51
VSS 61
VSS 83
VSS 94
VSS 107
VSS 120
VSS 130
VSS 31
VSSA
U1B
C1 2.2uF 16V
C2 2.2uF 16V
GND
GND
GND
VDDA/VREF+
AVCC
AVCC
C3 1uF 25V
C4 100nF 50V
GND
GND
VDD
+3.3V
C5 100nF 50V
C6 100nF 50V
C7 100nF 50V
C8 100nF 50V
C9 100nF 50V
C10 100nF 50V
C11 100nF 50V
C12 100nF 50V
C13 100nF 50V
C14 100nF 50V
C15 100nF 50V
C16 4.7uF 35V
GND
VBAT
+3.3V
C17 100nF 50V
GND
VDD33USB
+3.3V
+3.3V
C18 100nF 50V
C19 1uF 25V
GND
GND
Each power supply pair (VDD/VSS, VDDA/VSSA ...) must be decoupled with filtering ceramic capacitors as shown above. These capacitors must be placed as close as possible to, or below, the appropriate pins on the underside of the PCB to ensure good operation of the device. It is not recommended to remove filtering capacitors to reduce PCB size or cost. This might cause incorrect operation of the device.
Title
Size A4
Number
Revision
Date: 12-26-2024
Sheet of
File: C:\Users\..\MCU Power.SchDoc
Drawn By:



Title		
Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\MCU GPIO.SchDoc		Drawn By:



Title		
Size A4	Number	Revision
Date: 12-26-2024		Sheet of
File: C:\Users\..\MCU ADC.SchDoc		Drawn By: